

- [5] Zhang, C., Bengio, S., Hardt, M., Recht, B. & Vinyals, O. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM* **64**, 107–115 (2021).
- [6] Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [7] Brown, T. *et al.* Language models are few-shot learners. *Advances in neural information processing systems* **33**, 1877–1901 (2020).
- [8] Vaswani, A. *et al.* Attention is all you need. *Advances in neural information processing systems* **30** (2017).
- [9] Frank, M. C. Bridging the data gap between children and large language models. *Trends in Cognitive Sciences* (2023).
- [10] Tran, K., Bisazza, A. & Monz, C. The importance of being recurrent for modeling hierarchical structure. *arXiv preprint arXiv:1803.03585* (2018).
- [11] Linzen, T., Dupoux, E. & Goldberg, Y. Assessing the ability of lstms to learn syntax-sensitive dependencies. *Transactions of the Association for Computational Linguistics* **4**, 521–535 (2016).
- [12] Dehghani, M., Gouws, S., Vinyals, O., Uszkoreit, J. & Kaiser, L. Universal transformers. *arXiv preprint arXiv:1807.03819* (2018).
- [13] Weston, J. *et al.* Towards ai-complete question answering: A set of prerequisite toy tasks. *arXiv preprint arXiv:1502.05698* (2015).
- [14] Karim, M. & Rivera, S. Comparison of feed-forward and recurrent neural networks for bioprocess state estimation. *Computers & chemical engineering* **16**, S369–S377 (1992).

- [15] Sundermeyer, M. *et al.* Comparison of feedforward and recurrent neural network language models. In *2013 IEEE international conference on acoustics, speech and signal processing*, 8430–8434 (IEEE, 2013).
- [16] Alamia, A., Gauducheau, V., Paisios, D. & VanRullen, R. Comparing feed-forward and recurrent neural network architectures with human behavior in artificial grammar learning. *Scientific reports* **10**, 22172 (2020).
- [17] Elman, J. L. Finding structure in time. *Cognitive science* **14**, 179–211 (1990).
- [18] Servan-Schreiber, D., Cleeremans, A. & McClelland, J. L. *Encoding sequential structure in simple recurrent networks* (Carnegie Mellon University, Computer Science Department, 1988).
- [19] Tani, J. & Ito, M. Self-organization of behavioral primitives as multiple attractor dynamics: A robot experiment. *IEEE Transactions on Systems, Man, and Cybernetics-Part A: Systems and Humans* **33**, 481–488 (2003).
- [20] Namikawa, J. & Tani, J. Building recurrent neural networks to implement multiple attractor dynamics using the gradient descent method. *Advances in Artificial Neural Systems* **2009** (2008).
- [21] Wolpert, D. M. & Kawato, M. Multiple paired forward and inverse models for motor control. *Neural networks* **11**, 1317–1329 (1998).
- [22] Tani, J. & Nolfi, S. Learning to perceive the world as articulated: an approach for hierarchical learning in sensory-motor systems. *Neural Networks* **12**, 1131–1141 (1999).
- [23] Haruno, M., Wolpert, D. M. & Kawato, M. Mosaic model for sensorimotor learning and control. *Neural computation* **13**, 2201–2220 (2001).

- [24] Elman, J. L. *Rethinking innateness: A connectionist perspective on development*, vol. 10 (MIT press, 1996).
- [25] McClelland, J. L., Rumelhart, D. E., Group, P. R. *et al. Parallel distributed processing, volume 2: Explorations in the microstructure of cognition: Psychological and biological models*, vol. 2 (MIT press, 1987).
- [26] Hoffmann, J. *et al.* Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556* (2022).
- [27] Wang, Z., Ma, Y., Liu, Z. & Tang, J. R-transformer: Recurrent neural network enhanced transformer. *arXiv preprint arXiv:1907.05572* (2019).
- [28] Bulatov, A., Kuratov, Y. & Burtsev, M. Recurrent memory transformer. *Advances in Neural Information Processing Systems* **35**, 11079–11091 (2022).
- [29] Nishimoto, R., Namikawa, J. & Tani, J. Learning multiple goal-directed actions through self-organization of a dynamic neural network model: A humanoid robot experiment. *Adaptive Behavior* **16**, 166–181 (2008).
- [30] Radford, A., Narasimhan, K., Salimans, T., Sutskever, I. *et al.* Improving language understanding by generative pre-training (2018).
- [31] Brohan, A. *et al.* Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817* (2022).
- [32] Kingma, D. P. & Ba, J. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [33] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I. & Salakhutdinov, R. Dropout: a simple way to prevent neural networks from overfitting. *The journal of machine learning research* **15**, 1929–1958 (2014).